



# Liquid fuels demand and supply model

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## Manish handover

6 October focus | Source and reconcile the data

6 October 2026

Internal working discussion

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# Source the data before quantifying the fuel levers

Today: integrity, traceability, provincial update and a matched fuel balance

| Priority                      | Manish does                                                                                          | Return for review                                                                                |
|-------------------------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>1 Data integrity</b>       | Investigate incomplete provincial totals, duplicate refinery/history keys and unsupported baselines. | Flag; old value; source page/cell; proposed value; reason; resolved or open.                     |
| <b>2 Traceability</b>         | Trace priority data from publisher to original file, extract and consuming function.                 | Exact paths; API/download/manual; units; coverage; used, staged or reporting-only.               |
| <b>3 Provincial update</b>    | Verify newer observed provincial petrol/diesel sales, then seek 2025 and latest 2026 coverage.       | Province/product/period; observed versus estimated; national tie; missing periods.               |
| <b>4 Production and trade</b> | Reconcile production, imports, exports and stock movements for the same product and period.          | Matched litres; source definitions; stock treatment; unexplained residual.                       |
| <b>5 Driver evidence</b>      | Source agriculture, manufacturing, mining, passenger/freight, power, EV and plant series.            | Original evidence and extracts for each chart; revisions and coverage. Scenario settings follow. |

Ways of working: Nigel on main; Manish on a named branch from current origin/main. Nigel reviews changes before integration.

# Open the exact files and follow their data trail

All paths are relative to your clone of liquid\_fuels\_model

| Start here                  | Exact path beneath the project folder                                                                                     | What to check                                                                                                         |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Audit and direction</b>  | output/delivered/<br>Liquid_fuels_source_audit_2026_10_05.xlsx<br>source_profile_2026_10_05.html                          | Direction, Provincial gaps, Repeated keys; input access and consuming functions.                                      |
| <b>Provincial demand</b>    | assumptions/2026/timeseries/<br>fuel_sales_department_by_province.csv<br>fuel_sales_department_by_province_quarterly.csv  | Incomplete quarters and annual totals; source_file identifies the original workbook.                                  |
| <b>Production and trade</b> | assumptions/2026/timeseries/<br>energy_balance_department.csv<br>fuel_trade_fiasa.csv<br>fuel_trade_department_review.csv | Same product/period/units; 2024 diesel imports: 14.793 versus 10.8 bn litres remains a source disagreement.           |
| <b>Duplicate records</b>    | assumptions/2026/timeseries/<br>refinery_production.csv<br>historical_demand.csv                                          | 558 extra refinery rows and 15 extra historical rows.<br>refinery_production.csv feeds utilisation, despite its name. |
| <b>Baseline parameters</b>  | assumptions/2026/<br>vehicles.yaml; agriculture.yaml<br>industrial.yaml; marine.yaml                                      | Source of mileage, fuel mix, baselines and elasticities. Activity indices cannot silently replace parameters.         |

Raw = original PDF, Excel or API response in external/. Extract = CSV in assumptions/. Calculation = src/lfm/model/.

## Run one fetch command for each priority dataset

Use the staging setup on the next page before running these commands

```
$py = '.\.venv\Scripts\python.exe'
```

| Dataset    | PowerShell command                                     | Main extract                                         |
|------------|--------------------------------------------------------|------------------------------------------------------|
| Department | & \$py -m lfm.scripts.fetch_energy_dept --vintage 2026 | Sales, provinces, balances, prices                   |
| FIASA      | & \$py -m lfm.scripts.fetch_fuel_sales --vintage 2026  | Sales, imports and exports                           |
| Economy    | & \$py -m lfm.scripts.fetch_economy --vintage 2026     | GDP, population, Treasury; Stats SA workbook on disk |
| NaTIS      | & \$py -m lfm.scripts.fetch_natis --vintage 2026       | Fleet stock and registrations                        |
| naamsa     | & \$py -m lfm.scripts.fetch_naamsa --vintage 2026      | BEV, PHEV, hybrid and market sales                   |
| Eskom      | & \$py -m lfm.scripts.fetch_eskom --vintage 2026       | Eskom and IPP OCGT generation                        |

### Combined refresh: & \$py -m lfm.scripts.refresh\_sources --vintage 2026

Runs the six fetchers plus ACSA. Logs failed commands and compares sales; it does not collect every missing source. Append --offline only to reread originals already on disk. Capture each exit code and warnings.

# Keep candidate refreshes separate and test each source

Candidate files first; reviewed promotion into the vintage follows later

## PowerShell setup

```
$stage = "runs/manish_fetch_$(Get-Date -Format yyyyMMdd_HH:mm:ss)"  
New-Item -ItemType Directory "$stage/assumptions" -Force | Out-Null  
Copy-Item assumptions/2026 "$stage/assumptions/2026" -Recurse  
$oldA = $env:LFM_ASSUMPTIONS_DIR  
$oldD = $env:LFM_DATA_DIR  
$env:LFM_ASSUMPTIONS_DIR = (Resolve-Path "$stage/assumptions").Path  
$env:LFM_DATA_DIR = Join-Path (Get-Location) "$stage/data"  
New-Item -ItemType Directory "$stage/data/raw" -Force | Out-Null
```

## Stats SA manual prerequisite

Download the GDP Time series workbook into \$stage/data/raw/statssa/ before the economy command. The reader uses the Annual sheet; quarterly 2026 extraction needs support. Existing raw caches may be copied into the candidate raw folder.

## Checks and restoration

Compare old/new values, dates, units, duplicates and completeness. Preserve prior data on failed or partial fetches. Add tests for new parser support.

### Restore in finally if scripted:

```
$env:LFM_ASSUMPTIONS_DIR = $oldA  
$env:LFM_DATA_DIR = $oldD
```

### Then run:

```
& $py -m lfm check --vintage 2026  
& $py -m pytest -q
```

**A successful command is not numerical verification or adoption.**

## Return five distinct work packages for review

Checklist | Manish owns evidence collection and checks; Nigel reviews integration

| Work package                 | Checklist: required evidence                                                                        | Completion test                                                  |
|------------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| <b>[ ] 1 Integrity</b>       | Resolve provincial completeness/ties and duplicate refinery/history keys against original cells.    | Flag log: old/new value, reason; resolved or open with owner.    |
| <b>[ ] 2 Traceability</b>    | Map publisher -> original -> extract -> consuming function; list originals absent from main.        | Exact paths and links; units, dates and engine-use status.       |
| <b>[ ] 3 Demand update</b>   | Collect petrol/diesel sales by province after 2022; verify 2024 chart evidence and seek 2025/2026.  | Province/product/period; national tie; observed/estimated; gaps. |
| <b>[ ] 4 Fuel balance</b>    | Match actual production, imports, exports and stock movements. Resolve competing trade definitions. | Same-year/product litres; balance residual explained or open.    |
| <b>[ ] 5 Driver evidence</b> | Complete the seven driver rows on the next page. Add and test missing retrieval/parsers.            | One source record per series; coverage and revisions explicit.   |

**Today: evidence and tests. After review: Nigel integrates approved changes and updates the SCR pack; lever quantification follows.**

[Open detailed instructions and source links](#)

Source: 6 October agreed handover; source profile and audit; candidate data remain subject to review.

## Collect driver evidence with one checklist row per category

Data first | Record each row as sourced, partial or open; do not infer fuel litres from activity

| Driver             | Collect / public source                                                                    | Keep explicit                                        |
|--------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------|
| Passenger vehicles | NaTIS passenger stock/registrations; petrol/diesel split, mileage and efficiency evidence. | ICE stock split and mileage may remain open.         |
| Freight            | NaTIS goods-vehicle stock; Stats SA P7162 road/rail payload and available tonne-km.        | One revision vintage; tonnes are not tonne-km.       |
| Agriculture        | Stats SA P0441 agricultural activity; sector diesel-use/intensity evidence separately.     | National activity; no provincial litres inferred.    |
| Industry           | Stats SA P3041.2 manufacturing and P2041 mining output; separate series.                   | Indices measure output, not fuel consumption.        |
| Power generation   | Eskom and IPP OCGT generation; reported diesel burn where available.                       | FY versus CY; GWh and litres stay separate.          |
| Electrification    | naamsa BEV, PHEV and conventional hybrid new sales; fleet-stock evidence if available.     | Sales are a flow; HEV/PHEV still use fuel.           |
| Economic context   | Stats SA real GDP; department petrol/diesel prices, latest monthly observations.           | Real versus nominal; no elasticity calibrated today. |

For every series: original file + source URL/table + extracted CSV + period/units + revision check. Missing evidence: owner and next retrieval action.

# Return sourced data and explicit gaps

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Review evidence before setting lever  
scenarios

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